OFM Riaz Rahman Aranya

Ph.D. Student in Computer Science
Vision and Immersive Realities Lab (VIRLab)
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RESEARCH INTERESTS

- Medical Image Segmentation, Vision-Language Models for Clinical Applications, Medical Image Reconstruction, Explainable AI
- Radiology Report Generation, Longitudinal Change Analysis, Semi-Supervised Learning
- Synthetic-to-Real Domain Adaptation, Cross-Modal Image Reconstruction

EDUCATION

Ph.D. in Computer Science — **Expected Grad: 2027**
The University of Texas at San Antonio, San Antonio, TX
CGPA: 4.00/4.00 | Advisor: Dr. Kevin Desai

M.Sc. in Computer Science — **2026**
The University of Texas at San Antonio, San Antonio, TX
CGPA: 4.00/4.00 | Advisor: Dr. Kevin Desai

B.Sc. in Computer Science and Engineering — **2020**
Khulna University of Engineering & Technology, Khulna, Bangladesh
CGPA: 3.28/4.00

PUBLICATIONS

Journal Papers

1. A. Bagherian, **O.F.M. R. R. Aranya**, A. Kosub, et al. "Comprehensive Benchmarking of Deep Learning Approaches for Automated Astrocyte Segmentation in Traumatic Brain Injury." *Journal of Neuropathology and Experimental Neurology*, 2025. DOI
2. P. Saha, M. S. Sadi, **O.F.M. R. R. Aranya**, S. Jahan, F. Islam. "COV-VGX: An Automated COVID-19 Detection System Using X-ray Images and Transfer Learning." *Informatics in Medicine Unlocked*, 26:100741, 2021. DOI

Conference Papers

3. **O.F.M. R. R. Aranya**, P. Rad, K. Desai. "GRCD: Grounded Region Change Detection for Multi-Finding Chest X-Ray Pairs." *British Machine Vision Conference (BMVC 2026).* [arXiv] [Project] [Code]
4. M. B. Akram, A. H. M. N. Sakib, **O.F.M. R. R. Aranya**, R. Wijewickrama, K. Desai, M. Jadliwala. "ThermalTap: Passive Application Fingerprinting in VR Headsets via Thermal Side Channels." *Privacy Enhancing Technologies Symposium (PETS 2027)* (Submitted). [arXiv]

5. **O.F.M. R. R. Aranya**, K. Desai. "SRA-Seg: Synthetic to Real Alignment for Semi-Supervised Medical Image Segmentation." *International Conference on Pattern Recognition (ICPR 2026)*. [Paper] [arXiv] [Project] [Code]
6. **O.F.M. R. R. Aranya**, K. Desai. "TRACE: Temporal Radiology with Anatomical Change Explanation for Grounded X-ray Report Generation." *International Conference on Pattern Recognition (ICPR 2026)*. [Paper] [arXiv] [Project] [Code]
7. **O.F.M. R. R. Aranya**, K. Desai. "To Agree or To Be Right? The Grounding-Sycophancy Tradeoff in Medical Vision-Language Models." *CVPR 2026 Workshop (Med-Reasone)*. [Oral] [Best paper award] [PDF] [arXiv] [Project] [Code]
8. A. N. M. Sakib, **O.F.M. R. R. Aranya**, K. Desai, Z. Zhang. "Toward Faithful Segmentation Attribution via Benchmarking and Dual-Evidence Fusion." *CVPR 2026 Workshop (HOW)*. [PDF] [arXiv] [Code]
9. S. Das, **O.F.M. R. R. Aranya**, N. N. Labiba. "Brain Tumor Classification Using Convolutional Neural Network." *IEEE ICASERT*, 2019. DOI

RESEARCH EXPERIENCE

Graduate Research Assistant | *VIRLab, UTSA* **Jan 2026 – Present**
Advisor: Dr. Kevin Desai

- Research on medical image segmentation, cross-modal reconstruction, and vision-language models for clinical applications.
- Key results: SRA-Seg (89.34% Dice, ACDC, 10% labels), TRACE (90.2% grounding accuracy), GRCD (substantially outperforming TRACE across all clinical metrics).
- Discovered the grounding-sycophancy tradeoff in medical VLMs (CSI < 0.35 across all models).
- Clinical collaborations: OCT-to-histology cardiac reconstruction with a practicing cardiologist; cross-institutional neuropathology segmentation.

TEACHING EXPERIENCE

Instructor **Aug 2023 – Dec 2025**
Department of Computer Science, The University of Texas at San Antonio

- CS 2113 – Fundamentals of Object-Oriented Programming (Fall 2025)
- CS 3443 – Application Programming (Fall 2024)

Teaching Assistant
Department of Computer Science, The University of Texas at San Antonio

- CS 4843 – Cloud Computing (Summer 2025, Spring 2025)
- CS 3853 – Computer Architecture (Summer 2024)
- CS 3443 – Application Programming (Spring 2024)
- CS 2233 – Discrete Math Structures (Fall 2023)

INDUSTRY EXPERIENCE

Software Engineer | *RedDot Digital Ltd, Dhaka, Bangladesh* **Mar 2023 – Jul 2023**
Designed solution architecture and developed RESTful APIs for enterprise applications.

Jr. Software Engineer | *Nascenia Ltd, Dhaka, Bangladesh* **Dec 2020 – Feb 2023**
Full-stack developer (Ruby on Rails). Designed solution architectures, delivered multiple production systems, and mentored junior engineers.

HONORS & AWARDS

- **Best Paper Award**, Med-Reasoner Workshop, CVPR 2026.
- Published work (COV-VGX) recognized under the **United Nations Sustainable Development Goal 3** (Good Health and Well-being).
- Selected among top 4 in UTSA for Google fellowship candidate.
- 1st Runner Up, LICT Project Showcasing, Jashore IT Park, 2019.
- Board Scholarship (General Merit), Jashore Board, 2015.
- Champion, High Jump, KUET (2019, 2020); 2nd Runner Up, Long Jump (2019).

ACADEMIC SERVICE

Peer Review

- IEEE Signal Processing Letters (12 paper reviewed)
- Med-Reasoner Workshop, CVPR 2026 (2 papers reviewed)
- Med-Reasoner Workshop, NeurIPS 2026 (1 papers reviewed)
- SN Computer Science (1 paper reviewed)
- Informatics in Medicine Unlocked, Elsevier (1 paper reviewed)

LEADERSHIP & SERVICE

- Bangladesh Student Association at UTSA – **Treasurer** (2024–2025). Managed finances and event budgeting for 100+ member community.
- Jashore Association of KUET – **Vice President** (2019–2020), **General Secretary** (2018–2019).
- HACK (Hardware Acceleration Club of KUET) – **General Secretary** (2019–2020).
- Dream (Voluntary Blood Donation Society of KUET) – **Asst. General Secretary** (2019–2020).

PROFESSIONAL MEMBERSHIP

- IEEE Member (ID: 99988407), June 2024 – Present.
- CVF Member (ID: b7b14c38cc), March 2024 – Present.

TECHNICAL SKILLS

Languages: Python, Ruby, Java, C, C++
ML/DL: PyTorch, TensorFlow, HuggingFace Transformers
Web: Ruby on Rails, JavaScript, HTML/CSS
Tools: Docker, Git, Linux, LaTeX, MySQL